

Diagnosing Small-Model Failures in MemGPT-Style Memory Retrieval

Tool-Call Control, Query Policy, and Teacher-Trajectory Distillation

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Memory has to outlive the context window

A persistent assistant must answer about things no longer in context:

“What artist did I say I could get into?”

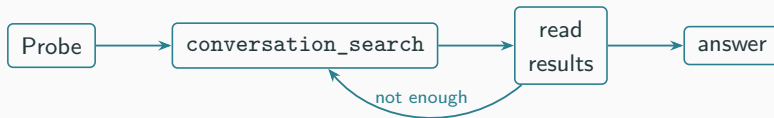
The answer lives in a **past session**—one specific earlier utterance, not world knowledge.

Speaker: “A little bit. I can get into Taylor Swift.”

The agent has to **recover that utterance** and ground the answer in it.

MemGPT makes recall a control problem

The model is a **controller over memory tools**—it does not answer at once.



The hidden difficulty is not the answer—it is **deciding how to operate the memory system**.

One question behind a low score

When a small open model is weak at deep memory retrieval, is it because

Knowledge

it cannot form the answer
even *with* the evidence?

Behavior

it cannot *operate* the
memory loop to get there?

We answer this by taking the loop apart, **one component at a time**.

Setup

- **Task:** MSC Deep Memory Retrieval, in a local Letta/MemGPT loop.
- **Models:** Llama-3-8B (student, LoRA target), Mistral-7B (frozen replay), GPT-4.1 (teacher & judge).

The twist: a literal substring retriever

A query returns past messages that *contain it as a substring*. audio studio location → nothing. studio, California → hit.

A good query is a **literal phrase likely to appear in past dialogue**, not a semantic summary.

How we test it: a diagnostic ladder

Each rung removes or repairs *one* part of the loop, so the next failure reads narrowly.

Intervention	What it isolates
Raw vanilla	can it enter the loop at all?
Strict template	just a formatting barrier?
Full history	can it answer without searching?
Teacher-trace replay	can it answer with the right evidence?
LoRA + query hint	does better query selection help?
Skeleton + reranking	can query policy be isolated and fixed?

Findings

They fail before retrieval even starts

- Raw small models: 0 / 5 loops — they never satisfy the tool-call contract.
- A strict formatting adapter restores the loop: 481 / 500 completed...
- ... yet containment is only 0.1954.

Fixing the *format* is not fixing the *retrieval*.

Remove the search, and it jumps

Put all past sessions directly in context — no tool calls, no retrieval:

containment 0.1954 → 0.5080

Most of the early loss is the model driving retrieval, not the model answering.

Our teacher: strong, but not gold

We distill GPT-4.1 traces — but the teacher is imperfect too.

- 398 / 500 traces judged correct...
- ... yet only 206 / 500 contain the exact reference.

We keep judge-approved traces and audit them — supervision, *not* gold policy.

Given the evidence, they answer fine

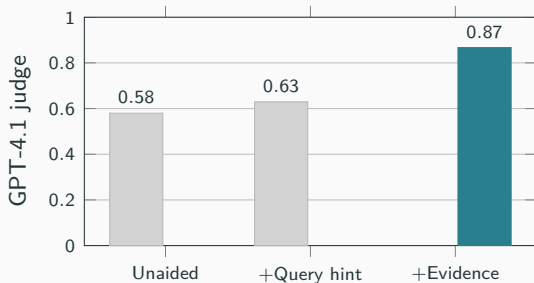
Frozen students, replayed the teacher's **retrieved evidence** (answer only):

GPT-4.1 judge	
Llama-3-8B	0.73
Mistral-7B	0.88

Mistral **fails the tool-call gate entirely**, yet answers well from evidence.

Answer generation is **not** the wall.

The bottleneck is getting evidence, not using it



Same 398 rows.

- Teacher **evidence**: +29 pts.
- Teacher **query hint**: +5 pts.

Handing over the evidence recovers the model; handing over the query strings barely moves it.

Query strings matter — but hints don't reach them

Replayed deterministically, **teacher queries** retrieve the answer-bearing memory far more often (searched subset):

teacher 0.48 vs. student 0.27 (retrieved-reference)

Where the student goes wrong

Reference: *"I have a cat."*

Teacher queries: pets, cat, cats

Student queries: pet, dogs, dogs $\Rightarrow$ answers *"I have a dog."*

Copying the probe, guessing a wrong prior, missing the discriminative literal — good queries exist, the model just doesn't **pick** them.

Memorizing queries fails; search-time control works

Hard-set SFT / DPO

fix a few hard rows,
hurt the full distribution.

No general query policy.

Candidate reranking

propose several queries,
keep the best by local feedback.

25/72 hard cases recovered; first to lift the full set.

Use the retriever as feedback at search time — don't bake teacher queries into weights.

Takeaways

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- A small memory agent fails **in stages**: contract → retrieval → grounding.
- Tool-call control, query policy, and evidence grounding are **different problems** — one teacher trajectory copies all three too coarsely.
- Build and evaluate small memory agents **component by component**.
- Put query-policy diagnostics **before** broad claims about long-term memory.

Thank you

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